Data storage for scientific computing

scicomp.aalto.fi/data/

Our brains aren't evolved to comprehend terabytes of data hidden in filesystems. You need to actively think about its arrangement.

The fundamental choices:

	Personal • Deleted when you leave Aalto • Ownership isn't shared	**Shared** • Stays with group when members depart • Access control by group
Backed up • Backups, snapshots, and disaster recovery	**Aalto home directories** Linux servers and desktops.	**Teamwork / "Department project"** Request from IT Services or your department.
Not backed up • You delete it and it's gone • Huge space available • High performance for analysis • Available on the HPC cluster	**Triton personal ("work")** Default 200 GB. We recommend Triton Scratch for most projects, but this gets you started.	**Triton project ("scratch")** 100s of GBs – 10s TB+ Main place for large computation data. We can make it as big as you need.
Cloud (via Aalto account) • Easy to share/edit with anyone • Not suitable for large-scale analysis • Expensive and non-sovereign	**OneDrive** **Aalto Google Drive** Better than consumer cloud but lost when you leave.	**Microsoft Teams groups** **Aalto Google Drive Shared Drives** Team-shared with multiple owners, preserved as people leave.

You should decide this together with your group: talk to your supervisor and colleagues and reuse the systems your group has already set up. When working for your group or supervisor, one would usually pick shared storage.

More details:

Name	Typical sizes	Backup and snapshots	Sharing	When you leave Aalto	Triton	Typical use	Notes
Teamwork	10-100s GB (up to TBs)	yes (same backup system)	yes	stays	login node	Large, backed up group space common outside SCI	IT Services manages this
Department project/archive	10s-100s GB		yes	stays	login node	Daily work	
Aalto Work	10s of GB		yes	stays	-		Access control only by Aalto org structure
Aalto homes	200 GiB		no	deleted	-	Own misc work	
Triton project ("scratch")	GBs-TBs	no (same filesystem)	yes	stays	all nodes	Big data or data being analyzed on Triton	Triton team manages this
Triton work	100s GB+		no	deleted	all nodes		
Triton homes	20 GiB	backups	no	deleted	all nodes	Small codes and configuration	Not high-performance.
Local SSD disks on Triton	TBs	no	no	deleted	some nodes	High-IO data processing	Available on some Triton nodes

SciComp Garage
scicomp.aalto.fi/help/garage/

There are too many options, it's OK to ask for help.
Come talk to us.
Online office hour, daily, 13:00: scicomp.aalto.fi/help/garage/

Aalto Science-IT Web: scicomp.aalto.fi Email: scicomp@aalto.fi Chat: scicomp.zulip.cs.aalto.fi

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Notes: our terminology

We have too many different names that are too similar to each other. As Aalto unifies we should work out consistency. A proposal for discussion is below (edit or make comments):

System	Canonical name	Other names	Deprecated names	Comments
/m/dept/project/X	Department project	Project, {CS,NBE,...} project, Department teamwork		
/m/dept/archive/X	Department archive	Archive, {CS,NBE,...} archive, Department teamwork		
Triton /scratch/dept/project/	Triton project	Triton scratch	"scratch"	
Triton /scratch/work/username/	Triton personal	Triton work	"work"	
Triton homes /home/username/	Triton home		"home"	
Teamwork (Aalto managed)	Teamwork	Aalto teamwork "Project"		
Aalto home directories	Aalto home			
Aalto work	Aalto work drive	"work"		The full name "Aalto work drive" makes the distinction from other works clear.